

A Hierarchical Approach for Region-Based Image Retrieval*

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Abstract – *In this paper, we propose a hierarchical approach for region-based image retrieval, which is based on wavelet transform for its decomposition property similarity with human visual processing. First, automated image segmentation is performed fast in the Low-Low(LL) frequency subband of wavelet transform which shows desirable low resolution of image. In the proposed system, boundaries between segmented regions are deleted to improve the robustness of region-based image retrieval against uncertainty of segmentation. Second, region feature vector is hierarchically represented by information in all wavelet subbands and each feature component of a feature vector is a combined color-texture feature. Such feature vector captures the distinctive feature(e.g., semantic texture) inside one region finely. Through experiment results and comparison with other methods, the proposed method shows good tradeoff between retrieval effectiveness and efficiency as well as easy implementation for region-based image retrieval.*

Keywords: Region-based image retrieval(RBIR), wavelet transform, coarse segmentation, hierarchical feature vector

1 Introduction

With the overwhelming growth of image databases, efficient image coding, manipulation, indexing and retrieval are required for effectively managing large image databases and make images easily accessible[1]. Content-based image retrieval (CBIR), querying images from large archives using image own content, has then become an important and hot issue in recent decades. Early CBIR methods exploit low-level image features such as color, texture and shape as keys to retrieve images[2, 3].

However, for the human visual system(HVS), the basic elements to carry semantic information are image regions which correspond to natural objects such as food, horse and flower if image segmentation is ideal. According to psychological research on the HVS, humans can effortlessly partition the semantic objects even with

ambiguous boundary in an image and recognize them with the interaction of fundamental visual components such as color and texture[8]. As a result, region-based image retrieval methods(RBIR), which segment images into regions and retrieve images based on the similarity between regions, have been investigated extensively for an attempt to bridge the gap between low image features and high image semantic concepts[4, 5, 6].

Semantic analysis and representation of image content is crucial for region-based image retrieval. Although the existing retrieval methods have improved the performance of RBIR, many of them still have one or more following limitations.

Many RBIR methods cannot provide desirable low-resolution image for image segmentation. Since the amount of raw image pixels is often very large, for simplifying data and deleting redundant details, a number of image retrieval systems either divide an image into unit pixel blocks whose averaged values are regarded as the pixel features or quantize the color space of an image into limited number of bins. Both methods are mathematically similar to image sub-sampling and rescaling, which cannot provide naturally low-resolution image for further image segmentation if an image contains high-frequency information such as sharp color variations.

Concerning region representation, a number of image retrieval approaches derive global region characterization. For example, the color/texture histogram or the center of the feature vector set in one region is often taken as the region feature vector. However, such global region description lacks local details or has limited discriminating power because much useful information is lost during the process that transforms the feature vector set to one single feature vector; It is also observed that color and texture features in region feature vector are commonly considered as separated ones, which is difficult and complex to tune the importance between two unrelated features in the distance functions for region matching.

In this paper we propose a hierarchical approach for region-based image retrieval. Our goal is to tackle above problems and improve image understanding and repre-

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sentation for RBIR. Our retrieval system is based on wavelet transform due to the similarity of its decomposition property with human visual processing.

First, automated image segmentation is performed in Low-Low(*LL*) frequency subband of image wavelet transform which shows desirable low-resolution image. Such coarse segmentation in *LL* subband is very fast with a small number of samples. Employing *LL* subband for data simplification is more natural and effective than blocking or color quantization. Furthermore, the boundaries between segmented regions are deleted in the proposed method, which makes region description more accurately and improves the robustness of RBIR against uncertainty of segmentation. Second, region feature vectors are hierarchically represented by the information in all wavelet subbands which depict detailed image content with different spatial-frequency resolution, and each feature component of feature vector is a combined color-texture feature. In contrast to global region description, the hierarchical feature vector captures visual distinctiveness of regions much finely. Note that the concept of coarse segmentation and fine region identification is very close to the human visual system, in which humans first partition an image into a few perceptual salient regions and then recognize them with fine visual feature(e.g.,semantic texture) inside the regions[8].

This paper is organized as follows. In section 2, we explain coarse image segmentation and hierarchical extraction of region feature vector. Section 3 introduces region-based image similarity measurement. The experiment results are discussed in section 4, and the last section is conclusion.

2 Coarse segmentation and fine region identification in wavelet domain

Wavelet transform utilizes series of filters with different frequencies and scales to process image and provides more perceptual filtering results. With proper level decomposition of an original image, the Low-Low(*LL*) frequency subband preserves image basic content(coarse configuration) and the other high-frequency subbands characterize image variations in different directions(texture details of image)[9].

In this paper, Daubechies wavelets [10] are selected because of their good localization properties and their efficient separation between low frequency and high frequency. According to our experimental experience, for the images with size of 384×256 or 256×384 pixels in our image database, a 3-level wavelet transform is feasible for desired division between basic image content and complement details. That is, the decomposition level $L=3$ in this paper.

An image is decomposed into multiple subbands with different frequencies by image wavelet transform,

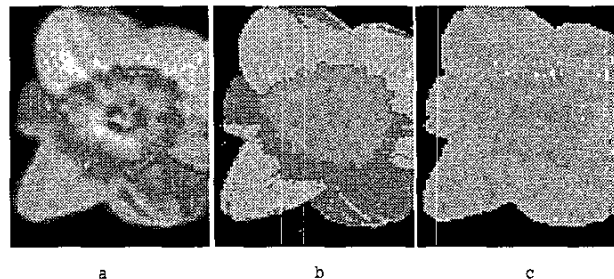


Figure 1: An example of segmentation results in the *LL* subband. Fig.a is an original image, Fig.b is segmentation result with $K=4$. Fig.c is segmentation result with optimal $K=3$. We enlarge the results to view conveniently.

Coarse image segmentation is performed in the *LL* subband, which reduces the amount of raw pixel data significantly while preserving essential information needed for image understanding task. To identify segmented regions exactly, detailed color and texture information is required. Therefore, region feature vectors are extracted from all different frequency subbands to describe region content finely in our method. In the following, section 2.1 proposes coarse segmentation in the *LL* subband and section 2.2 describes our fine region feature vector.

2.1 Coarse segmentation without boundary

Fair image segmentation desires low frequency resolution to ignore relatively small regions in an image. The *LL* subband in wavelet domain has been proven to provide naturally coarse resolution.

In this paper unsupervised K -means clustering algorithm [11] is applied in the *LL* subband for fast coarse segmentation. *HSV* color space is chosen in our method since it is similar to human perception and then can reflect actual perceptual color distance[12, 13].

To reflect color and proper texture information, six features are used for clustering. Three features are three color components of wavelet coefficients in the *LL* subband. In order to obtain the other three, first, for each point in the *LL* subband, its corresponding points in the same spatial location at the neighboring HL_3, LH_3, HH_3 subbands(referring to fig.2) are localized, and then the *H* color components of them are taken as the rest three features of the clustered feature vector. The motivation for using these three features is that they characterize texture properties in high frequency to some extent[14].

The cluster number K is critical for the semantic effectiveness of image segmentation. The number of K is automatically determined in our implementation. Suppose the set $\{\mu_i, i = 1, \dots, Q\}$ of Q points is to be

clustered into K clusters with centers $\{\hat{\mu}_j, j = 1, \dots, K\}$. In our method, we iterate K from minimum cluster number $K=2$ to maximum cluster number $K=12$. The “optimal” K is determined automatically if it minimizes *Cluster Dispensation Value (CDV)* defined as:

$$CDV = \frac{1}{K} \sum_{j=1}^K R_{C_j} \cdot \sum_{\mu_i \in C_j} D(\mu_i, \hat{\mu}_j) \quad (1)$$

Where $D(\mu_i, \hat{\mu}_j)$ is the Euclidean distance between μ_i and the center $\hat{\mu}_j$ of the j -th cluster C_j . R_{C_j} is the area of the minimum-enclosing rectangle(MER) of C_j . We can see that too clumped or too scattered segmentation would make the *CDV* value large in equation 1. We do not apply the clustering algorithm to the perceptually homogeneous images, that is, $K=1$. In our image retrieval system, one image is considered as uniform pattern if mean square deviation of H color values in the whole image is smaller than a threshold($\zeta=0.08$). The maximum value $K=12$ shows that an image is assumed to have less than 12 different objects. Usually, for practical images, the image segmentation produces much less number of objects. The maximum cluster number is rarely met. In figure.1 one segmentation result is shown. Fig.1.c shows the optimal segmentation result of cluster number $K=3$ with minimized *CDV* value. Compared with this, in fig.1.b where $K=4$, the computed *CDV* value is larger. It can be observed that the pixel clustering result is a little scattered.

Image segmentation still keeps a difficult problem now for creating an intelligent algorithm whose performance is close to the human visual system. Humans can identify distinct objects in an image and give meaningful assignment of the pixel points in the image. Although those pixel points cannot be assigned unambiguously to objects, human visual system always performs well[19]. Motivated by this observation, in this paper, the boundary points between the segmented regions that can be defined as a discontinuity in the vector field representing color image function [12] are first localized in the clustered LL subband, and then those boundary points with their 3×3 neighborhood are deleted. The advantage of deleting ambiguous information of boundary in an image is that it can improve the robustness of the image retrieval system against segmentation-related uncertainty. Furthermore, the feature vector of the segmented region can be extracted more exactly without including those boundary points.

2.2 Fine region identification

Only information in low-frequency subband would not be enough for effective region identification, which is known in computer vision[19]. Compared with coarse segmentation, for region identification, fine visual feature such as detailed texture inside one region is desired, which is characterized by high-frequency information.

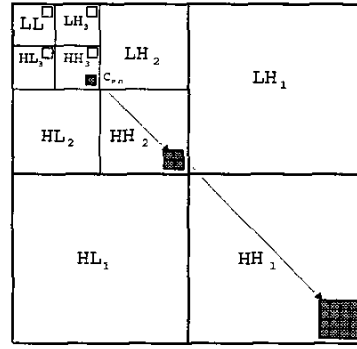


Figure 2: Spatial orientation tree in wavelet domain. A wavelet coefficient $C_{m,n}$ and all its descendents are shown in the colored parts. The arrow points from the subband of the “parents” to the subband of the “children”. For one point in the LL subband, its corresponding points in the similar spatial location at the HL_3 , LH_3 , HH_3 subbands are also shown.

In a hierarchical subband system, a tree structure, called spatial orientation tree, defines the spatial relationship between coefficients in low-frequency subband and coefficients in high-frequency subband, which is presented initially in [15, 16]. Referring to fig.2, at the hierarchical pyramid, with the exception of the highest frequency subbands, every coefficient at a given frequency subband can be related to a set of coefficients at the next higher frequency subband of similar orientation. The coefficients at a low frequency subband are called “parents”, and all coefficients corresponding to the same spatial location at the next frequency subbands of similar orientation are called “children”. With the exception of the lowest frequency subband LL , one “parents” in a certain frequency subband has a group of 4 adjacent pixels as “children” in the next higher frequency subband. For example, in fig.2, for one point $C_{m,n}$ in HH_3 subband, its “children” in similar orientation HH_2 subband are as:

$$Children(C_{m,n}) = \{C_{2m,2n}, C_{2m,2n+1}, C_{2m+1,2n}, C_{2m+1,2n+1}\} \quad (2)$$

For one region r_i in an original image, due to the wavelet decomposition property, all its sub-regions in the wavelet domain are different spatial-frequency resolution of the region r_i . The sub-region in the LL subband shows coarse content of r_i and the sub-regions in the other high-frequency subbands show complementary texture details of r_i . It can be expected that the combination of all sub-regions will describe r_i content much finely. Referring to fig.3, for one segmented region LLr_i in the LL subband, its corresponding sub-regions in the same spatial location at the LH_3, HL_3, HH_3 subbands are first localized. Then according to the spatial ori-

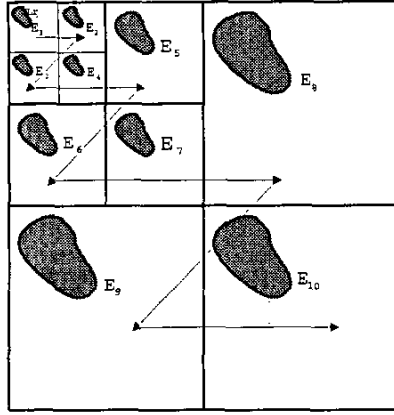


Figure 3: The illustration for the generation of one hierarchical region feature vector. For each sub-region in the wavelet domain, its weighted sum E_p of local energy in three color channels is computed as one color-texture feature component in the region feature vector. All these feature components are ordered as the arrow scan direction and constituted as one region feature vector

entation tree structure, their “children” regions are localized at the other high-frequency subbands. Noticing that since the boundaries of the segmented regions in the LL subband have been deleted, the sub-regions in the other frequency subbands also do not include boundaries. Without ambiguous semantic interference of boundary, the feature vector of the region r_i in the original image can be characterized more accurately.

In general, wavelet coefficients obtained by decomposing an image are not invariant with respect to translation, rotation and scaling of an image. In our method, the energy of wavelet coefficients is utilized to define the region feature vector. The energies of wavelet coefficients in the high frequency subbands have been proven to be effective for discerning texture in [14]. For each sub-region in the wavelet domain, its weighted sum of local energy in three color channels is computed, which is invariant with respect to image translation or rotation. Such weighted sum of local energy can be expected to reflect the color-texture feature of one sub-region and be regarded as one feature component. As shown in fig.3, the feature components in all the subbands are ordered as the sequence to generate a region-based feature vector. The local energy E_p of one sub-region is calculated by the following function:

$$E_p = w_H \cdot E_{pH} + w_S \cdot E_{pS} + w_V \cdot E_{pV}, p = 1, 2, \dots, P \quad (3)$$

Where E_{pH}, E_{pS}, E_{pV} are three color local energy components, $P = 3 \cdot L + 1$, P is the total number of subbands with the wavelet decomposition level L . It also means that the region feature vector in our approach is

P -dimensional. With $L = 3$ in this paper, the region feature vector in our indexing algorithm is 10-dimensional. For three color components H, S, V , the hue shows more importance than the saturation and the value for human visual perception[12]. This can be achieved by setting the weight for the hue channel to a higher value than the other weights. In our implementation, the weight proportion set as $2 : 1 : 1 (w_H : w_S : w_V)$ in equation 3 emphasizes the importance of the hue, which is also used in [13].

E_{pH} is calculated in equation 4. E_{pS} and E_{pV} are also defined as the similar way.

$$E_{pH} = \sum_{m,n \in S} (C_{m,n}^H)^2 \cdot K(2^{-l}(m_0 - m), 2^{-l}(n_0 - n)) \quad (4)$$

Here, $C_{m,n}^H$ is the H color component of wavelet coefficient $C_{m,n}$ in wavelet domain. S indicate one sub-region in wavelet subband. l indicates which decomposition level the sub-region is in. K is smoothing filters and the Gaussian filter is a desirable choice. (m_0, n_0) denotes the sub-region center. [18] indicates that squaring in combination with a logarithmic nonlinearity normalizing is the best combination for the nonlinearity to describe texture feature. At last we obtain the hierarchical region feature vector FV^{r_i} for one region r_i in an image as:

$$\begin{aligned} FV^{r_i} &= (fv_1^{r_i}, fv_2^{r_i}, \dots, fv_P^{r_i}) \\ &= (\log E_1, \log E_2, \dots, \log E_P) \end{aligned} \quad (5)$$

The advantages of our region feature vector are listed below:

1. It characterizes region content finely since it includes all different spatial-frequency information of region r_i .
2. The weights for feature components in the distance function can be tuned flexibly and meaningfully when measuring the similarity between two regions in the next section since the feature components are consistent color-texture features in wavelet domain.

3 Region-based image similarity comparison

Proper weight adjustment for feature components of region feature vector is important when measuring the similarity between two regions with a distance function. In some existing image retrieval systems, each feature component in a region feature vector may be a separated feature. It is difficult to determine exactly how much more important one color component is than another texture component. A particular feature component may be more important than another component in one image, while in another image the opposite may be true.

In our image retrieval system, each feature component in the region feature vector is a consistent color-texture feature presented by wavelet coefficients that characterize the region content in certain spatial-frequency resolution, therefore, the weight values in the distance function can be tuned more easily and meaningfully.

The distance function between two regions q_i, t_j is first defined as equation 6. It is known that even if given unlimited viewing time, humans will not scan all areas of a scene, but will instead attend to several perceptual interest regions which continually attract human attention[17]. Inspired by this observation and wavelet transform property, the weight proportion is set as the values in table 1, where those feature components in the second and the third wavelet decomposition levels are given more importance than those in the first level, and the weight for the component in the LL subband is set larger than the others. The intuition of such weight initialization is that the main color-texture information of the visually significant objects in an image is focused on relatively low frequency subbands.

$$\begin{aligned}
Dis(q_i, t_j) = & W_3[w_1(fv_1^{q_i} - fv_1^{t_j})^2 + \\
& w_2(fv_2^{q_i} - fv_2^{t_j})^2 + \\
& w_3(fv_3^{q_i} - fv_3^{t_j})^2 + \\
& w_4(fv_4^{q_i} - fv_4^{t_j})^2] + \\
& W_2[w_5(fv_5^{q_i} - fv_5^{t_j})^2 + \\
& w_6(fv_6^{q_i} - fv_6^{t_j})^2 + \\
& w_7(fv_7^{q_i} - fv_7^{t_j})^2] + \\
& W_1[w_8(fv_8^{q_i} - fv_8^{t_j})^2 + \\
& w_9(fv_9^{q_i} - fv_9^{t_j})^2 + \\
& w_{10}(fv_{10}^{q_i} - fv_{10}^{t_j})^2] \quad (6)
\end{aligned}$$

Table 1: The weights used in the distance function

$W_3 = 3$	$w_1 = 2$
	$w_2 = 1$
	$w_3 = 1$
	$w_4 = 1$
$W_2 = 3$	$w_5 = 1$
	$w_6 = 1$
	$w_7 = 1$
$W_1 = 1$	$w_8 = 1$
	$w_9 = 1$
	$w_{10} = 1$

The overall image similarity between one query image Q with regions $\{q_1, \dots, q_i, \dots, q_M\}$ and one image T with regions $\{t_1, \dots, t_j, \dots, t_N\}$ in the database is measured as a weighted sum of the similarity between region pairs, which is defined in equation 7 and 8.

$$Sim(Q, T) = \sum_{i=1}^M \theta_i \cdot Sim(q_i, T) \quad (7)$$

$$Sim(q_i, T) = \sum_{j=1}^N \lambda_{i,j} \cdot e^{-Dis(q_i, t_j)} \quad (8)$$

M, N is the number of segmented regions in the query image Q and the image T in the database, respectively. θ_i is the importance weight for region q_i in query image Q . It is initialized to $\frac{1}{M}$ and will be updated according to user's feedback, which will be described in the next section. The parameter $\lambda_{i,j} \in (0, 1]$ is chosen to adjust the effect of region q_i and t_j on the similarity measure. In the current implementation of the proposed method, the effect of region size is considered. $\lambda_{i,j}$ is defined as:

$$\lambda_{i,j} = \frac{size(q_i) + size(t_j)}{size(Q) + size(T)} \cdot \frac{size(q_i)/size(Q)}{size(t_j)/size(T)} \quad (9)$$

Here, $size$ is defined as the number of pixels inside one region. Parameter $\mu_{i,j}$ takes into account the relative size of the regions with respect to the images they belong to, which is based on the consideration that larger regions are more likely to attract human attention than smaller ones[17], and the term $\nu_{i,j}$ takes into account the similarity of size ratios of regions, which is derived from the fact that similar objects may have similar size ratios in the compared images.

4 Experiment

In this section, the retrieval performance (retrieval effectiveness and efficiency) of the proposed method will be evaluated through experiments.

The image database is consisted of 10,000 color photographs with size of 384×256 or 256×384 pixels, covering a wide range of semantic categories, from natural scenery to artificial objects. The image database is gathered from the well-known Corel image collection, which is used for the evaluation of many existing image retrieval systems[6, 7]. Classification of the image database into semantic categories is determined by the domain specialists, and reflects the human perception of image similarity. There are 100 images in each semantic category.

This system is developed in Visual C++6.0 language and implemented on PC with CPU Pentium4, 2.2G/Windows2000, RAM 512M. Its interface is shown in fig.4.

As a evaluation method, the retrieval precision is most frequently used in the current image retrieval systems. For one query image, a retrieved image is considered as a relevant one if and only if it belongs to the same image category as the query image. Based on this, retrieval precision is defined as the percentage of retrieved relevant images in the top- N retrieved images.

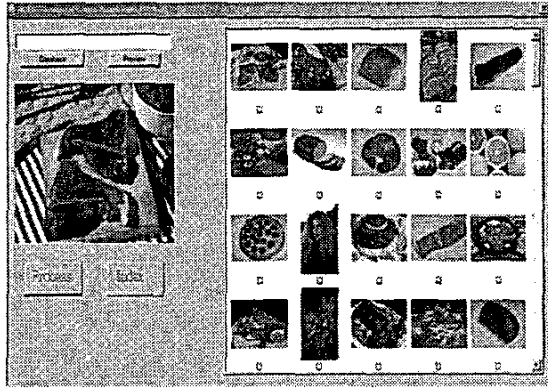


Figure 4: User Interface of the proposed retrieval system. The larger image on the left is query image. The check box below each retrieved image will be labeled if it is considered as a relevant image by the user

The effectiveness of our hierarchical region feature vector is evaluated through comparison with three image retrieval methods.

The first method is the WBIIS(Wavelet-Based Image Indexing and Searching) system[2]. WBIIS is selected for comparison because it also extracts feature vectors from wavelet coefficients and is one of the representative methods that exploit low image features for indexing. In WBIIS, an image is first decomposed by a 4-level wavelet transform, then those wavelet coefficients in the four lowest frequency subbands(LL, HL_4, LH_4, HH_4) are stored as image feature vector. The information in the other high frequency subbands is not taken into consideration in WBIIS.

The second one is the SIMPLicity(Semantic-Sensitive Integrated Matching for Picture Libraries)[7], which is one of the state-of-the-art region-based image retrieval methods in recent years. In the SIMPLicity system, the region feature vector is represented globally by the averaged value of the feature vector set in one region, which is a 6-dimensional vector, three components for color and three components for texture.

The third retrieval method is designed to prove that our hierarchical region feature vector has better discriminating power than a global region feature vector represented by color/texture histogram, which is denoted as GlobalColor. In GlobalColor, image segmentation is performed by EM algorithm and in each region color histogram with 218 bins is extracted globally as region feature vector. Note that the image segmentation and representation in GlobalColor is similar to that of Blobworld[6]. At last the image similarity is computed by equation 7- 9.

In total, there are four image retrieval methods implemented in this experiment. In order to provide numerical evaluation results, we select 10 semantic image

categories(each includes 100 images) as the query set that are shown in table 3. The image retrieval is carried out in the whole image database of 10,000 images. The retrieval precision is calculated as the averaged value in the top 30 images. The results of category-wise retrieval precision of the four methods are summarized in fig.5. The numbers in x-axis correspond to the image categories in table 3.

As shown in fig.5, for those relatively smooth images including large homogeneous regions such as “Weather” and “Mountain”, all the four methods perform the image retrieval equally well. However, for those image categories that include regions with distinctive color and texture features such as “Foods” and “Zebra”, our image indexing method outperforms the other three methods significantly. It is also noticed that for “Africa” and “Beach” image classes, our method does similar or a little worse performance than SIMPLicity and GlobalColor because most images in these two classes do not have objects with clear semantic concepts but just include many excessively small regions.

In all, as fig.5 summarizes, WBIIS algorithm shows the worst performance in the four retrieval methods; SIMPLicity and GlobalColor provide better retrieval results than WBIIS, which demonstrates that region-based retrieval methods could be more powerful than the retrieval schemes employing low image features. SIMPLicity outperforms GlobalColor mainly due to the lack of texture features in GlobalColor method. Our retrieval method gives best performance than the other methods for most semantic image categories and brings in average 18.5%, 6.2%, 7.8% increase in retrieval precision compared to WBIIS, SIMPLicity, GlobalColor methods respectively. The reason is explained explicitly as the follows:

- In WBIIS system, since the feature vectors are only consisted of the wavelet coefficients in low frequency subbands, much useful information in high frequency subbands that means important perceptual details is discarded.
- In GlobalColor, the region content is globally characterized by color histogram, which cannot reflect texture details and spatial information. Besides, it is also proved through this experiment that the retrieval performance degrades as the number of dimensions increases(*the curse of dimensionality*). In GlobalColor method, a high 218-dimension feature vector is used.
- In SIMPLicity, since the region feature vectors are represented by the averaged value in one region, they have limited discriminating power since much semantic information is lost during the averaging process. Besides, it is difficult and complex to decide the weights in the distance function for the separated color and texture features.

Table 2: Image categories in the second experiment

Number	Image Category
1	Weather
2	Mountain and glaciers
3	Foods
4	Flowers
5	Zebra
6	Buildings
7	Buses
8	Eagles
9	Africa people and village
10	Beach

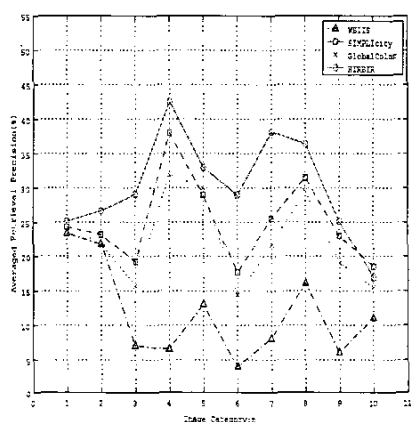


Figure 5: The category-wise image retrieval precision of our method(denoted as HIRBIR), compared with WBIIS, SIMPLicity and GlobalColor methods. The numbers in x -axis correspond to the image categories in table 3.

- In the proposed method, the hierarchical feature vectors extracted from all different frequency subbands can capture semantic details inside one region. Furthermore, the adjustment of weights in region matching functions in our method is more easily and meaningfully because each component of our feature vector is a consistent color-texture feature.

5 Conclusions

A novel hierarchical RBIR method is proposed in this paper. The advantages of our method is summarized as:

- Coarse segmentation is conducted in the LL sub-band fast, which significantly lowers the computation cost while keeping essential information needed for image segmentation task.

- Region feature vector is extracted hierarchically from all different wavelet frequency subbands, which captures the distinctive features inside one region finely.
- With our hierarchical feature vector, the distance functions for region matching are tuned meaningfully and easily.

The experiment results verify the effectiveness and efficiency of the proposed method. Our indexing system yields better retrieval accuracy with the image categories that include objects with distinctive color and texture features. Our system also can retrieve images in which an identical object is located with various translations.

In the future work, we want to refine our image segmentation method for more semantic-meaningful description of image content. Other information of images such as shape and context will be considered to enhance our retrieval system. The relevance feedback mechanism will be considered, in which the weights for feature vector and region will be tuned on-line to capture user's intention better.

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